Project: Worked examples
Title: Shaft under combined bending and torsion
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the bending and torsional stresses in a solid round shaft, with stress-concentration factors, and the factors of safety against yield. It covers static loading only, not fatigue or hollow shafts.

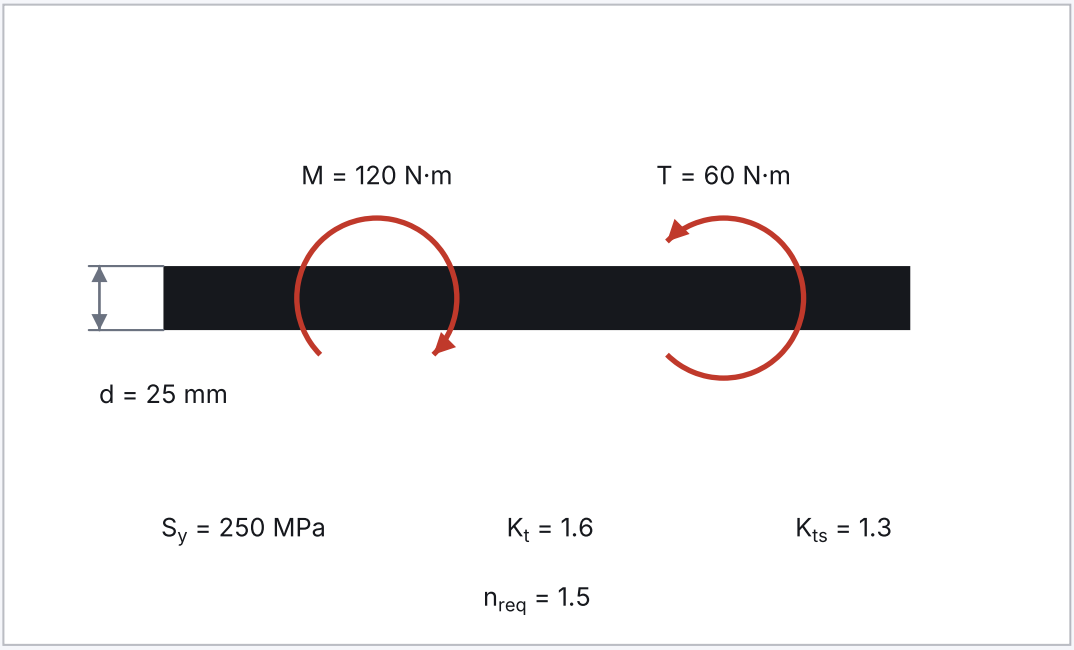


Figure 1 — A solid round shaft under bending and torsion at the section. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Bending stress, with its stress concentration	$\sigma = K_t \cdot 32M/(\pi d^3)$
Torsional shear stress, with its stress concentration	$\tau = K_{ts} \cdot 16T/(\pi d^3)$
Maximum shear stress (Tresca)	$\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)}$
von Mises equivalent stress	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$
Factors of safety against yield, the lower governing	$n_T = (S_y/2)/\tau_{max}$; $n_{VM} = S_y/\sigma'$

Basis: Maximum-shear-stress and distortion-energy theories[1](§5-4 and §5-5, 10th ed.); Shaft design for stress[1](§7-4, 10th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Shaft diameter	25	mm	drawing: DRG-SH-402 rev A	1 … 2000 mm
S_y	Yield strength	250	MPa	drawing: DRG-SH-402 rev A	50 … 3000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
M	Bending moment (magnitude)	120	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m
T	Torque (magnitude)	60	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m
K_t	Bending stress-concentration factor (1 for none)	1.6		spec: SPEC-MECH-03 design criteria	1 … 5
K_{ts}	Torsional stress-concentration factor (1 for none)	1.3		spec: SPEC-MECH-03 design criteria	1 … 5
n_{req}	Required factor of safety	1.5		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. The shaft is a solid round section, linear elastic, under static loading. (The stress formulas of the method assume this.)

4. Working

[8] $\sigma = K_t * 32 * M / (pi * d^3)$ [1](§7-4)

$= 1.6 \times 32 \times 120\ N\cdot m / (\pi \times (25\ mm)^3)$

$= 125.16\ MPa$

Bending stress

[9] $\tau = K_{ts} * 16 * T / (pi * d^3)$ [1](§7-4)

$= 1.3 \times 16 \times 60\ N\cdot m / (\pi \times (25\ mm)^3)$

$= 25.424\ MPa$

Torsional shear stress

[10] $\tau_{max} = sqrt((\sigma / 2)^2 + \tau^2)$ [1](§5-4)

$= sqrt((125.16\ MPa / 2)^2 + (25.424\ MPa)^2)$

$= 67.549\ MPa$

Maximum shear stress (Tresca)

[11] $\sigma_{vm} = sqrt(\sigma^2 + 3 * \tau^2)$ [1](§5-5)

$= sqrt((125.16\ MPa)^2 + 3 \times (25.424\ MPa)^2)$

$= 132.68\ MPa$

von Mises stress

[12] $n_T = (S_y / 2) / \tau_{max}$

$= 250\ MPa / 2 / 67.549\ MPa$

$= 1.8505$

Factor of safety, maximum shear stress

[13] $n_{VM} = S_y / \sigma_{vm}$

$= 250\ MPa / 132.68\ MPa$

$= 1.8842$

Factor of safety, distortion energy

[14] $n_{min} = min(n_T, n_{VM})$

$= min(1.8505, 1.8842)$

$= 1.8505$

Governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
τ_{max}	67.55 MPa	Maximum shear stress
σ_{vm}	132.7 MPa	von Mises stress
n_{min}	1.8504	Governing factor of safety
σ	125.2 MPa	Bending stress
τ	25.42 MPa	Torsional shear
n_T	1.8504	Factor of safety (Tresca)
n_{VM}	1.8841	Factor of safety (von Mises)
u_n	0.81	Utilisation of factor of safety against the required factor (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	1.5	1.8505	0.81	[1](§7-4)

7. Applicability

None.

8. Verification

Template mech.shaft-stress@1.1.1 Content hash e68a699ed74d32f0 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.